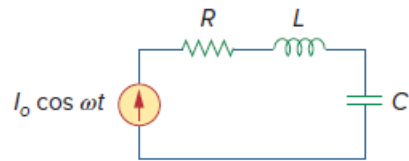


Problem

Compute the complex power supplied by the current source in the series RLC circuit in Fig. 11.87.

Figure 11.87:



Step-by-step solution

Step 1 of 1

Let, $S = S_R + S_L + S_C$

Where, $S_R = P_R + jQ_R$

$$S_R = \frac{1}{2} I_0^2 R + j0$$

$$S_L = P_L + jQ_L$$

$$S_L = 0 + j \frac{1}{2} I_0^2 \omega L$$

$$S_C = P_C + jQ_C$$

$$S_C = 0 - j \frac{1}{2} I_0^2 \frac{1}{\omega C}$$

Hence,

$$S = \frac{1}{2} I_0^2 \left[R + j \left(\omega L - \frac{1}{\omega C} \right) \right]$$